

Config-Driven RL for Just-in-Time Adaptive Interventions

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Problem Definition

Goal: Optimise personalised physical-activity interventions using RL.

MDP: $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$

- $\mathcal{S}$: 36-state factored MDP (step bin, sleep, day of week, burden)
- $\mathcal{A}$: 4 actions (idle, movement suggestion, goal reminder, journal)
- $\mathcal{P}$: Unknown — estimated via LLM simulation or hand-curated tables
- $\mathcal{R}$: $\alpha \cdot f(\text{step}) + (1-\alpha) \cdot g(\text{sleep}) - \lambda \cdot \mathbb{I}[\text{action} \neq \text{idle}]$

Core challenge: How do we iterate on RL agents without live patients?

Approach: A config-driven simulation framework with LLM-bootstrapped user models.

Existing Work

	HeartSteps V2	PEARL	StepCountJITAI
Type	MRT	4-arm RCT	Simulator
Sample	$n \approx 97$	$n = 13,463$	Synthetic
Algorithm	TS bandit	RL bandit	DQN
Actions	~ 5 types	155 nudges	4 types
Outcome	78.4% improved	+296 steps*	DQN ≈ 3000
Theory	SDT / FBM	COM-B	—

*RL vs control at 1 month,

$$p = 0.0002.$$

Key gap: No existing framework combines *config-driven experiment definition* with *LLM-bootstrapped user simulation* for rapid agent prototyping.

Our Approach

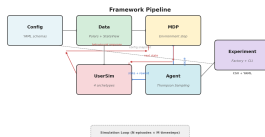
Two pillars:

① Config-Driven Framework

- Entire MDP defined in YAML — no source code changes between experiments
- Expression-based rewards, 3 transition types, 7 RL/bandit agents all registered by name
- A researcher can define state, actions, rewards, and transitions in one file

② Synthetic LLM Patients

- LLMs as transition models — sampled next-state probabilities from personas
- 5 behavioural archetypes: base, goal-driven, resistant, social responder, stable maintainer
- Plug in a new persona table, run the same experiment config



What We Have Done

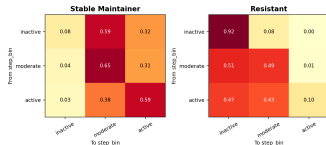
	MVP Simulator	LLM Bootstrap			Experiments
Purpose	Prove the pipeline	Generate	patient	be-	Evaluate agents
		haviour			
State space	2-state	36-state factored			Both
Transitions	Hand-curated rules	22,320 LLM prompts			From YAML config
Output	285 hyperparam combos	42 tables, 85 figures			7 personas $\times$ 7 agents
Cost	\$0	\$0.27 total			\$0

Key point: A 2-state hand-curated MDP and a 36-state LLM-bootstrapped MDP run through the *same* YAML schema, same agent library, same evaluation runner.

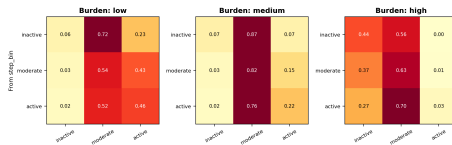
Initial Results

Transition analysis — persona differentiation is real, burden degrades intervention effectiveness.

Persona-Specific Transition Patterns Under Movement Suggestion
Averaged across all conditions



Movement Suggestion Effectiveness Degrades Under Higher Burden
Base Persona, averaged across time-of-day, day boundary, sleep



Persona	Optimal DP	HeartSteps (% opt)	Best Bandit (% opt)
Stable Maintainer	441.8	440.8 (99.8%)	402.5 (91.1%)
Base	379.7	379.1 (99.9%)	327.8 (86.3%)
Goal-driven	414.1	414.5 (100.1%)	323.0 (78.0%)
Initial DeepSeek	381.6	373.6 (97.9%)	297.5 (78.0%)
Initial GLM 5.2	374.6	374.0 (99.8%)	326.8 (87.2%)
Social Responder	313.5	306.8 (97.9%)	140.1 (44.7%)
Resistant	214.6	163.5 (76.2%)	151.7 (70.7%)

HeartSteps reproduced as a sanity check — 97–100% of optimal confirms the simulator is well-formed.

What is Next

① Sensor Foundation Models

- Wearable FM POC (NormWear, 768-dim embeddings) — feature branch
- FM-guided dynamic action spaces (15 interventions, 3 archetypes)

② Additional LLM Backends

- Beyond DeepSeek V4 Flash: GLM 5.2, Qwen, Claude
- Multi-turn persona dialogues for richer state transitions

③ Prompt Engineering

- COM-B behaviour-change taxonomy integration
- Structured JSON schema enforcement, chain-of-thought prompting

④ Synthetic Clinical Results

- Calibrate transition tables to real-world effect sizes (PEARL: +200–300 steps)
- Integration with wearable datasets (All of Us, UK Biobank)